

## LECTURE 15

### GROUP AND RING THEORY

We shall write  $M \otimes_R N$  for  $T(M, N)$  and  $m \otimes n$  for  $(m, n) + B(M, N) = \phi(m, n)$ , for  $m \in M$  and  $n \in N$ . Since every element of  $B(M, N)$  is mapped to zero under the canonical mapping  $\phi$ , we obtain the following:

**Theorem 11.7.** *Let  $M$  and  $N$  be right and left  $R$ -modules, respectively. Then*

- (i)  $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$ ,
- (ii)  $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$ ,
- (iii)  $mr \otimes n = m \otimes rn$ ,

for all  $m, m_1, m_2 \in M$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$ .

Next we prove:

**Theorem 11.8.** *Let  $M$  and  $N$  be right and left  $R$ -modules, respectively.*

- (a) *Let  $\lambda \in \mathbb{Z}$ ,  $m \in M$ , and  $n \in N$ . Then*

$$(\lambda m) \otimes n = \lambda(m \otimes n) = m \otimes \lambda n.$$

*In particular,*

$$0 \otimes n = 0 = m \otimes 0 \quad \text{and} \quad (-m) \otimes n = -(m \otimes n) = m \otimes (-n).$$

- (b) *Every element in  $M \otimes_R N$  is of the form  $\sum_{\text{finite}} m_i \otimes n_i$ , for  $m_i \in M$  and  $n_i \in N$ .*

*Proof.* (a) If  $n$  is kept fixed, the mapping given by  $m \mapsto m \otimes n$  is a group homomorphism of  $M$  into  $M \otimes_R N$ . (Handout) Indeed, let  $\theta_n : m \mapsto m \otimes n$ . Then  $\theta_n(\lambda m) = \lambda \theta_n(m) = \lambda(m \otimes n)$ . Thus for  $\lambda \in \mathbb{Z}$ ,

$$(\lambda m) \otimes n = \lambda(m \otimes n).$$

Similarly

$$m \otimes \lambda n = \lambda(m \otimes n).$$

- (b) Since every element of  $A(M, N)$  is of the form

$$\sum_{\text{finite}} \lambda_i(m_i, n_i), \quad \text{for } \lambda_i \in \mathbb{Z}, m_i \in M, n_i \in N,$$

it follows that every element of  $M \otimes_R N$  is a finite sum  $\sum \lambda_i(m_i \otimes n_i)$ . However,

$$\lambda_i(m_i \otimes n_i) = \lambda_i m_i \otimes n_i,$$

and so every element of  $M \otimes_R N$  is a finite sum of elements of the form  $m \otimes n$ , for  $m \in M$  and  $n \in N$ .  $\square$

We refer to an element of  $M \otimes_R N$  of the form  $m \otimes n$  as a *simple tensor* (or *elementary tensor*). By the above result, not every element of  $M \otimes_R N$  is a simple tensor.

**11.3. Module structure of the tensor product.** If  $M$  and  $N$  are right and left  $R$ -modules, respectively, over an arbitrary ring  $R$ , then  $M \otimes_R N$  is an abelian group. However, if  $R$  is commutative, then  $M \otimes_R N$  can be given the structure of an  $R$ -module. A natural definition for making  $M \otimes_R N$  a right  $R$ -module is  $(m \otimes n)r = mr \otimes n$ , for  $m \in M$ ,  $n \in N$ , and  $r \in R$ . It is however somewhat non-trivial to verify that this is indeed well defined. We proceed to show this, assuming  $R$  is a commutative ring.

Let  $r \in R$ . Consider the homomorphism

$$f_r : A(M, N) \longrightarrow M \otimes_R N$$

of the free abelian group  $A(M, N)$  to  $M \otimes_R N$  given by  $f(m, n) = mr \otimes n$ . We note that the elements of the form

$$(11.1) \quad (m_1 + m_2, n) - (m_1, n) - (m_2, n)$$

and

$$(11.2) \quad (m, n_1 + n_2) - (m, n_1) - (m, n_2)$$

map, respectively, to

$$(m_1 r + m_2 r) \otimes n - m_1 r \otimes n - m_2 r \otimes n = m_1 r \otimes n + m_2 r \otimes n - m_1 r \otimes n - m_2 r \otimes n = 0$$

and

$$mr \otimes (n_1 + n_2) - mr \otimes n_1 - mr \otimes n_2 = 0,$$

for all  $m, m_1, m_2 \in M$  and  $n, n_1, n_2 \in N$ . Furthermore, for all  $m \in M$ ,  $n \in N$ , and  $s \in R$ , the elements of the form

$$(11.3) \quad (ms, n) - (m, sn)$$

map to

$$(msr) \otimes n - mr \otimes sn = (msr) \otimes n - (mrs) \otimes n = 0,$$

since  $R$  is commutative.

Thus  $f_r$  vanishes on the subgroup  $B(M, N)$  of  $A(M, N)$  generated by elements of the form (11.1), (11.2) and (11.3). So  $f_r$  induces a homomorphism

$$\bar{f}_r : A(M, N)/B(M, N) = M \otimes_R N \longrightarrow M \otimes_R N,$$

where  $\bar{f}_r((m, n) + B(M, N)) = mr \otimes n$ , equivalently  $\bar{f}_r(m \otimes n) = mr \otimes n$ . This proves that given  $m \otimes n \in M \otimes_R N$  and  $r \in R$ , the element  $mr \otimes n$  is uniquely defined. Thus  $(m \otimes n)r = mr \otimes n$  is well defined, and it is easy to verify that  $M \otimes_R N$  becomes a right  $R$ -module. Incidentally,  $M \otimes_R N$  is a left  $R$ -module in a natural way. (Over a commutative ring  $R$  a right  $R$ -module can be naturally looked upon as a left  $R$ -module.)

**Standing assumption:** Generally when considering tensor products, one always assumes that  $R$  is with unity, which is what we will also do, albeit belatedly.

The next result makes explicit use of this assumption.

**Theorem 11.9.** *Let  $M$  be a right  $R$ -module. Then  $M \otimes_R R \cong M$  as additive abelian groups. Further, if  $R$  is commutative, then  $M \otimes_R R \cong M$  as  $R$ -modules.*

*Proof.* Consider a group homomorphism

$$f : A(M, R) \rightarrow M$$

given by  $f(m, r) = mr$ . Clearly  $B(M, R) \subseteq \ker f$  and so  $f$  induces a homomorphism  $\bar{f}$  from  $A(M, R)/B(M, R) = M \otimes_R R$  to  $M$  given by

$$\bar{f}(m \otimes r) = mr.$$

Also the map  $g : M \rightarrow M \otimes_R R$  given by  $g(m) = m \otimes 1$  is a group homomorphism whose inverse is  $\bar{f}$ . Thus  $\bar{f}$  is an isomorphism as desired. (Handout) In the case when  $R$  is commutative, then by the definition of scalar multiplication of elements of  $R$  with those of  $M \otimes_R N$ , we have

$$\bar{f}((m \otimes r)s) = \bar{f}(ms \otimes r) = (ms)r = m(sr) = m(rs) = (mr)s = \bar{f}(m \otimes r)s.$$

Thus  $\bar{f}$  is an  $R$ -homomorphism and is indeed an  $R$ -isomorphism as shown in the first case.  $\square$

One key application of tensor products is a more general version of the above: if  $M$  is an  $R$ -module and  $S$  is a commutative ring that contains  $R$ , then  $M \otimes_R S$  is an  $S$ -module. So the tensor product enables us to consider  $M$  as an  $S$ -module. We will not cover the proof of this fact.

Here are some other useful result.

**Theorem 11.10.** *Let  $M$ ,  $N$ , and  $P$  be  $R$ -modules over a commutative ring  $R$ . Then*

- (i)  $M \otimes_R N \cong N \otimes_R M$  as  $R$ -modules;
- (ii)  $(M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$  as  $R$ -modules.

*Proof.* First one needs to recall that if  $M$  is a right  $R$ -module over a commutative ring  $R$ , then  $M$  is a left  $R$ -module by defining  $rm = mr$ , for  $r \in R$  and  $m \in M$ . Consider the homomorphism  $f : A(M, N) \rightarrow N \otimes_R M$  given by  $f(m, n) = n \otimes m$ . Since  $B(M, N) \subseteq \ker f$ , the map  $f$  induces the homomorphism  $\bar{f} : M \otimes_R N \rightarrow N \otimes_R M$  given by  $\bar{f}(m \otimes n) = n \otimes m$ . One can, similarly, obtain a homomorphism from  $N \otimes_R M$  to  $M \otimes_R N$  which is the inverse of  $\bar{f}$ , proving that  $M \otimes_R N \cong N \otimes_R M$ . The proof of part (ii) is similar. [For example, for a fixed  $p \in P$ , show that there is an  $R$ -homomorphism  $M \otimes_R N \rightarrow M \otimes_R (N \otimes_R P)$  sending  $m \otimes n \mapsto m \otimes (n \otimes p)$ . Then consider the map  $(M \otimes_R N) \times P \rightarrow M \otimes_R (N \otimes_R P)$  which is defined using the previous homomorphisms (one for each  $p$ ).]  $\square$

**Example 11.11.** Let  $M$  be a right  $R$ -module and  $e$  an idempotent in  $R$ . Then  $M \otimes_R Re \cong Me = \{me \mid m \in M\}$  as groups. Indeed, the balanced map from  $M \times Re$  to  $Me$  defined by  $(m, re) \mapsto mre$  induces a homomorphism from  $M \otimes_R Re$  to  $Me$ . The homomorphism  $me \mapsto m \otimes e$  is the inverse to the previous homomorphism. (Note that if  $me = 0$ , then

$$m \otimes e = m \otimes e^2 = me \otimes e = 0 \otimes e = 0$$

too.

Also an alternative approach to proving the isomorphism is note that

$$M \otimes_R Re = (M \otimes_R R)e \cong Me.$$

where we have used Theorem 11.9 for the isomorphism.)

**11.4. Tensor product of homomorphisms.** Let  $M, M'$  be right  $R$ -modules and  $N, N'$  be left  $R$ -modules. Let  $f : M \rightarrow M'$  and  $g : N \rightarrow N'$  be  $R$ -homomorphisms. Then the mapping

$$A(M, N) \rightarrow M' \otimes_R N'$$

in which

$$(m, n) \mapsto f(m) \otimes g(n)$$

is a group homomorphism which vanishes on the subgroup  $B(M, N)$ ; note that it is a group homomorphism essentially immediately from how elements in  $A(M, N)$  are defined. Thus the above mapping induces a homomorphism from  $A(M, N)/B(M, N) = M \otimes_R N$  to  $M' \otimes_R N'$ , where the element  $(m, n) + B(M, N) = m \otimes n$  maps to  $f(m) \otimes g(n)$ . We denote this homomorphism as  $f \otimes g$ . Thus

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

The mapping  $f \otimes g$  is called the *tensor product of  $f$  and  $g$* . When  $R$  is commutative, the mapping  $f \otimes g$ , as defined above, becomes an  $R$ -homomorphism from the  $R$ -module  $M \otimes_R N$  to the  $R$ -module  $M' \otimes_R N'$ .

**Theorem 11.12.** *Let  $M, M', M''$  be right  $R$ -modules and  $N, N', N''$  be left  $R$ -modules. Suppose  $f : M \rightarrow M', f' : M' \rightarrow M'', g : N \rightarrow N',$  and  $g' : N' \rightarrow N''$  are  $R$ -homomorphisms. Then*

$$(f' \otimes g')(f \otimes g) = (f'f \otimes g'g).$$

*Further, if  $f$  and  $g$  are isomorphisms of  $M$  onto  $M'$  and  $N$  onto  $N'$ , respectively, then  $f \otimes g$  is also an isomorphism.*

*Proof.* The proof follows from the definition of the tensor product of homomorphisms.  $\square$

As an application of the above result, we prove the following.

**Theorem 11.13.** *Let  $(M_i)_{i \in I}$  and  $(N_j)_{j \in J}$  be families of right and left  $R$ -modules, respectively. Then*

$$\left( \bigoplus_{i \in I} M_i \right) \otimes_R \left( \bigoplus_{j \in J} N_j \right) \cong \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j),$$

*by the group isomorphism which maps*

$$\left( \sum_i x_i \right) \otimes \left( \sum_j y_j \right)$$

*onto*

$$\sum_{i,j} x_i \otimes y_j.$$

*Proof.* The map

$$f : \left( \bigoplus_{i \in I} M_i \right) \times \left( \bigoplus_{j \in J} N_j \right) \longrightarrow \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j)$$

given by

$$f \left( \left( \sum_i x_i, \sum_j y_j \right) \right) = \sum_{i,j} x_i \otimes y_j$$

is balanced and gives rise to a group homomorphism

$$\bar{f} : \left( \bigoplus_{i \in I} M_i \right) \otimes_R \left( \bigoplus_{j \in J} N_j \right) \longrightarrow \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j)$$

such that

$$\bar{f} \left( \left( \sum_i x_i \right) \otimes \left( \sum_j y_j \right) \right) = \sum x_i \otimes y_j.$$

On the other hand, the inclusions

$$\lambda_i : M_i \rightarrow \bigoplus_{k \in I} M_k \quad \text{and} \quad \mu_j : N_j \rightarrow \bigoplus_{\ell \in J} N_\ell$$

give rise to group homomorphisms

$$g_{ij} := \lambda_i \otimes \mu_j : M_i \otimes_R N_j \longrightarrow \left( \bigoplus_{k \in I} M_k \right) \otimes_R \left( \bigoplus_{\ell \in J} N_\ell \right).$$

There exists a unique group homomorphism

$$\bar{g} : \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j) \longrightarrow \left( \bigoplus_{i \in I} M_i \right) \otimes_R \left( \bigoplus_{j \in J} N_j \right)$$

which extends all the  $g_{ij}$ 's. Clearly  $\bar{f}$  and  $\bar{g}$  are inverses of each other and hence they are isomorphisms.  $\square$

The following weaker version of the final statement of Theorem 11.12 still holds.

**Theorem 11.14.** *Let  $M, M'$  be right  $R$ -modules and  $N, N'$  be left  $R$ -modules, where  $R$  is a commutative ring. Suppose  $f : M \rightarrow M'$  and  $g : N \rightarrow N'$  are  $R$ -homomorphisms. If  $f$  and  $g$  are surjective, then  $f \otimes g$  is also surjective.*

*Proof.* Since  $f \otimes g$  is an  $R$ -homomorphism, to show it is surjective, it suffices to show that every simple tensor in  $M' \otimes N'$  is in the image. For such a simple tensor  $m' \otimes n'$ , we can write  $m' = f(m)$  and  $n' = g(n)$ , since  $f$  and  $g$  are surjective. Therefore  $m' \otimes n' = f(m) \otimes g(n) = (f \otimes g)(m \otimes n)$ .  $\square$

However, the same does not hold for the property of being injective. Here is an example.

**Example 11.15.** Let  $R = \mathbb{Z}$  and let  $\alpha : \mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{Z}/p^2\mathbb{Z}$  be multiplication by  $p$ , so  $\alpha(x) = px$ . This is injective. Now if we tensor  $\mathbb{Z}/p\mathbb{Z}$  with itself, and consider the tensor product of the identity map with  $\alpha$ :

$$1 \otimes \alpha = \mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p^2\mathbb{Z}$$

we observe that

$$a \otimes x \mapsto a \otimes px = pa \otimes x = 0,$$

so  $1 \otimes \alpha = 0$ .

The above also links to the fact that the tensor product of submodules may not be a submodule of the tensor product:

**Example 11.16.** As  $p\mathbb{Z} \cong \mathbb{Z}$  as abelian groups, via the map  $pn \mapsto n$ , we have (see also Exercise Sheet 15)

$$\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} p\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}$$

as abelian groups, via the mappings

$$a \otimes pn \mapsto a \otimes n \mapsto na \pmod{p}.$$

Therefore  $1 \otimes p$  in  $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} p\mathbb{Z}$  is non-zero, since the above isomorphism identifies it with 1 in  $\mathbb{Z}/p\mathbb{Z}$ . However  $1 \otimes p$  in  $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$  is 0, since  $1 \otimes p = p \otimes 1 = 0 \otimes 1 = 0$ . (This computation does not work in  $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} p\mathbb{Z}$  since we cannot move  $p$  to the left of  $\otimes$  and leave 1 behind, as  $1 \notin p\mathbb{Z}$ .)

Here  $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} p\mathbb{Z}$  is not a subgroup of  $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ , even though  $p\mathbb{Z}$  is a subgroup of  $\mathbb{Z}$ .

This brings us to the end of Group and Ring Theory. The course Representation Theory of Finite Groups uses the concepts of modules and tensor products, and the courses Infinite Groups and Galois Theory do make use of the group theory developed here.